

Ch # 23

"Gauss Law"

1, 4, 6, 7, 23, 27, 33

Q1)

Given Data

$$E = 1800 \text{ N/C}$$

Square

$$a = \gamma = 3.2 \text{ mm}$$

Area = a^2

$$\phi = ?$$

$$\theta = 35^\circ$$

$$\theta = 180 - 35^\circ$$

Sol.

$$\phi = E \cdot dA$$

$$= E dA \cos \theta \Rightarrow 1800 (3.2 \times 10^{-3}) \cos 35^\circ$$

~~$$\phi = 0.01$$~~

$$\phi = -0.015 \frac{\text{Nm}^2}{\text{C}}$$

Q 4)

Given Data

$$E = 3.0 \text{ m N/C}$$

$$a = 11 \text{ cm} = 0.11 \text{ m}$$

$$\phi = ?$$

Area of

$$\text{circle} = \pi r^2$$

Sol

$$\phi = E \cdot dA$$

$$= E \pi r^2 \Rightarrow 3 \times 10^{-3} \times \pi \times (0.11)^2$$

$$\boxed{\phi = 1.13 \times 10^{-4} \text{ Nm}^2/\text{C}}$$

Net flux is opposite to the rim

$$\boxed{\phi = -1.13 \times 10^{-4} \text{ Nm}^2/\text{C}}$$

Q.6)

Given Data

$$Q_{net} = ?$$

$$\vec{E} = -34 \hat{k} \text{ N/C}$$

$$r = 3 \text{ m}$$

$$\vec{E} = 20 \hat{k} \text{ N/C}$$

Sol.

$$\phi_{net} = E \cdot \vec{r} - E \cdot \vec{r}$$

$$= -34 (3 \times 3) + 20 (-3 \times 3)$$

$$= -306 +$$

$$\phi_{net} = -486 \frac{\text{Nm}}{\text{C}}$$

$$E = \frac{F}{Q_0}$$

$$Q = \phi \epsilon_0$$

$$\epsilon_0$$

$$Q = \phi \epsilon_0 \Rightarrow -486 \times 8.85 \times 10^{-12}$$

$$= -4.3 \times 10^{-9}$$

$$Q = -4.3 \text{ nC} \quad \underline{\text{Ans}}$$

Q7)

Given Data

$$V = 1.8 \text{ HC}$$

$$r = 55 \text{ cm}$$

$$\phi = ?$$

Sol.

$$\phi = \frac{Q}{\epsilon_0} \Rightarrow \frac{1.8 \times 10^{-6}}{8.85 \times 10^{-12}}$$

$$= 0.2033 \times 10^{-6+12}$$

$$= 0.2033 \times 10^{-6+12}$$

$$\boxed{\phi = 2 \times 10^5 \frac{\text{N.m}^2}{\text{C}}}$$

$$\phi = 828$$

$$\phi = \frac{Q}{\epsilon_0}$$

$$Q = 7.3 \times 10^{-9}$$

Q 23)

Given Data

$$L = 42 \text{ cm} = 0.42$$

$$d = 12 \text{ cm} = 0.12$$

$$E = 2.3 \times 10^5 \text{ N/C}$$

$$Q_{\text{net}} = ?$$

Sol.

$$\sigma = \frac{Q}{A}$$

Area of
Drum

$\pi d L$

$$Q = \epsilon_0 E \pi d L$$

$$\Rightarrow 8.85 \times 10^{-12} \times 2.3 \times 10^5 \times 3.14 \times 0.12 \times 0.42$$

$$= 3.2 \times 10^{-7}$$

$$= 0.32 \times 10^{-6} \Rightarrow \boxed{0.32 \text{ HC}}$$

b)

$$L = 28 \text{ cm} = 0.28$$

$$d = 8 \text{ cm} = 0.08$$

$$Q_{\text{net}} = ?$$

Sol.

$$Q = \sigma A$$

$$= \epsilon_0 E \pi d L$$

$$= 8.85 \times 10^{-12} \times 2.3 \times 10^5 \times 3.14 \times 0.08 \times 0.28$$

$$\boxed{Q = 1.4 \times 10^{-7} \text{ C}}$$

Q.27

Given Data

Charge density $= \lambda = 3.6 \text{ nC/m}$

$$r = 1.5 \text{ cm} = 0.015$$

$$\lambda = \frac{q}{l}$$

$$q = \lambda l$$

$$\sigma = \frac{q}{A} \Rightarrow \frac{\lambda l}{2\pi r l}$$

$$\Rightarrow \frac{3.6 \times 10^{-9}}{2\pi \times 0.015}$$

$$= 0.0942$$

$$= 3.6 \times 10^{-9}$$

$$0.0942$$

$$\sigma = 3.8 \times 10^{-8} \text{ C/m}^2 \quad \underline{\text{Ans}}$$

33.

a) $\vec{E} = (\sigma / 2 \epsilon_0) (-\hat{j})$ right plate
 $+ (\sigma / 2 \epsilon_0) (-\hat{j})$ left plate
 $E = (\sigma + 2\sigma) (-\hat{j}) + 2(\sigma / 2 \epsilon_0) (-\hat{j}) =$

b) $\vec{E} = \overset{R}{(\sigma / 2 \epsilon_0) \hat{j}} + \overset{L}{(\sigma / 2 \epsilon_0) (-\hat{j})} = 0$

c) B/w plates

$$\vec{E} = \frac{\sigma}{2 \epsilon_0} (-\hat{j}) + \frac{\sigma}{2 \epsilon_0} (-\hat{j}) = \frac{\sigma}{\epsilon_0} (-\hat{j})$$

$$= - \left(\frac{7 \times 10^{-22}}{8.85 \times 10^{-12}} \right) \hat{j}$$

$$= - 7.91 \times 10^{-11}$$